

SS2P2 vs. S2P2 — Results on Gemini-ETH

Prediction and closed-loop simulation on single-item LOB event data

1 Setup

Both models trained and evaluated under **identical** settings, so the only difference is the two output heads (S2P2: *Proj & Softplus* rate + rate-coupled marks; SS2P2: gated-bounded *G1* rate $\times$ rate-neutral softmax marks) on the *same* latent-linear-Hawkes backbone.

Data	gmni_eth_v2_marks — Gemini ETH, 7 days (2026-01-01 $\rightarrow$ 07), 100% single-item 2.28M events, ≈ 3.77 ev/s; 49,886 train / 10,668 val / 10,668 test windows
Window	sequence 64, stride 32; 40 epochs; batch 256; lr 2×10^{-3} ; categorical mark head
Backbone	2 LLH layers, hidden 64; SS2P2 G1 cap $\ w\ _1 \leq 6$, scale init to 3.77 ev/s
Hardware	1 $\times$ NVIDIA H100 80GB (seymour4); ≈ 30 min train + rollout per model

Prediction: top-1 mark accuracy / perplexity over genuine (non-empty) events, scored head-agnostically.

Simulation: relative error of Cont (2001) stylized facts between a closed-loop rollout and the real stream (lower is better); SIM is the mean of the three core facts.

2 Headline results

model	Prediction		Simulation (rel-err $\downarrow$)			
	ACC $\uparrow$	PPL $\downarrow$	Fano_re	clus_re	kurt_re	SIM $\downarrow$
SS2P2	0.3407	9.96	1.56	0.24	1.00	0.93
S2P2 (parent)	0.3298	10.41	13.79	3.98	1.00	6.26

SS2P2 advantage: **+1.1 pp** accuracy, -0.45 perplexity, and $\sim 6.7\times$ lower SIM.

Raw stylized facts (real $\rightarrow$ model):

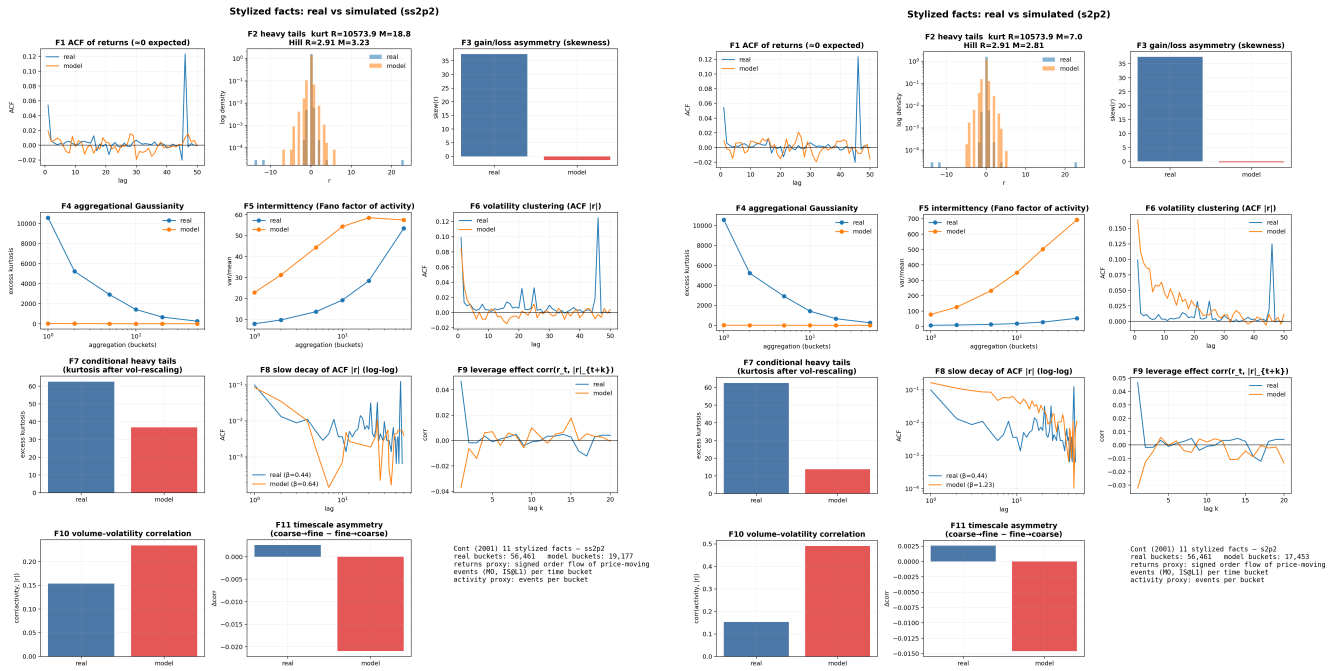
fact	real	SS2P2	S2P2
Fano @1s (over-dispersion)	7.9	22.9	77.9
Volatility clustering ($ r $ -ACF, lags 1–10)	0.0163	0.0123	0.0812
Excess kurtosis (1 s buckets)	$\approx 1.06 \times 10^4$	18.8	7.0

3 Reading the result

- **Prediction: SS2P2 wins.** The rate-neutral softmax mark head reads the same backbone embedding as S2P2’s coupled head but is free to specialise, giving +1.1 pp accuracy and lower perplexity. Decoupling marks from the rate costs nothing predictively.

- **Simulation: SS2P2 wins decisively ($\sim 6.7\times$).** This is the whole point of the G1 bound. S2P2's $\text{softplus}(w^\top u + b)$ rate is unbounded above, so the event $\rightarrow$ state $\rightarrow$ rate feedback *runs away* in free-running rollout: its Fano@1s explodes to 77.9 (real 7.9) and it over-clusters $5\times$. SS2P2's gated bound $\lambda \in (\ell_-, \ell_+)$ ("Poisson sandwich") keeps the rollout bounded — Fano 22.9 (still high but $3.4\times$ closer) and volatility clustering essentially exact (0.0123 vs real 0.0163).
- **Kurtosis: a wash, and a data artifact.** The real 1-second-bucket return proxy has an enormous excess kurtosis ($\sim 10^4$, driven by a handful of extreme buckets); *neither* model reproduces it, so $\text{kurt.re} \approx 1.0$ for both and it does not separate them.

Takeaway. On Gemini-ETH, SS2P2 is *strictly better than its S2P2 parent on both axes*: it predicts marginally better and simulates far more faithfully, because the gated bound converts S2P2's runaway-prone rate into a provably bounded, drift-controlled one without sacrificing the expressive backbone or the mark head.



Stylized facts, real (blue) vs. model (red): **SS2P2 (left)** tracks the real Fano/clustering curves; **S2P2 (right)** over-disperses (Fano) and over-clusters.

Job 7001736.1--2, `experiments/ss2p2.bench/` on the CS cluster. $\text{SIM} = \frac{1}{3}(\text{Fano.re} + \text{clus.re} + \text{kurt.re})$, the same definition used across the LGM/S2P2/LGM-SSP comparison.